MILITARY SPECIFICATION

PARALLEL, FREQUENCY-CODED FIRING SYSTEMS

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers the general requirements for a parallel, frequency coded firing system consisting of one encoder and one or more decoders. Procurement of system components requires additional data provided on component drawings listed in Data List DL 154AS100.

1.2 Classification - The parallel frequency coded-firing system is installed in an aircraft weapons-release circuit and consists of one encoder connected in parallel with from 1 to 15 decoders. These components are installed in weapon-release circuits of military aircraft to prevent accidental weapon release.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-W-5088	Wiring: Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-E-17555	Electronic and Electrical Equipment, Accessories and Repair Parts; Packaging and Packing of

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MIL-T-18303	Test Procedures: Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

STANDARDS

Military

MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods

DATA LISTS

Naval Air Systems Command

DL 154AS100	Frequency Coded Firing System (Parallel System)

DRAWINGS

Naval Air Systems Command

127AS130	Test Procedure, MIL-STD-704 Spike Voltage Test

2.1.1 Availability of Documents -

(1) When requesting specifications, standards and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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3. REQUIREMENTS

3.1 Quality Assurance - This specification makes provision for Quality Assurance testing.

3.2 Parts and Materials - In the selection of parts and materials, the requirements of MIL-E-5400 shall be complied with. Fulfillment of major design objectives shall be the prime consideration.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-E-5400.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship except as otherwise specified herein.

3.3.1 Reliability -

3.3.1.1 Operating Life - Each component of the parallel, frequency coded firing system shall be capable of operating 129,000 times during its operating life when tested and accepted as outlined under 4.5.

3.3.1.2 Reliability in Mean Number of Firings Between Failure (MFBF) - The capability of each component of the parallel, frequency-coded firing system to operate 108,000 times between failures shall be demonstrated as outlined under 4.4.3.

3.3.2 Cabling and Connections - The equipment shall provide for the use of cables and connectors in accordance with MIL-E-5400.

3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements, and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.3 Interchangeability - The equipment shall meet the interchangeability requirements of MIL-E-5400.

3.3.4 Interference Control - The generation of radio interference by each component of the frequency-coded firing system and the vulnerability of the system to radio interference shall be limited to the requirements of MIL-STD-461 for Class ID Equipment. Transient loading conditions shall be exempt from the test requirements.

3.3.5 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.3.6 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions, and for making laboratory bench tests.

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Temperature	Room ambient (30 ± 10°C)
Altitude	Test Facility ground level
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	27.5 ± 0.5 VDC

3.3.7 Service Conditions - Encoders shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 2 equipment except as modified herein. Decoders shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 3 equipment except as modified herein.

3.3.7.1 Vibration - Frequency-coded firing systems shall operate satisfactorily when subjected to the vibration requirements of Curve AT of MIL-STD-810. All equipment except the encoders shall be subjected to the additional requirement of 0.20 inch double-amplitude between the frequency range of 5 to 15 Hertz.

3.3.7.2 Temperature - Unless otherwise specified by the encoder or decoder individual drawing, the maximum operating temperature for encoders shall be in accordance with MIL-E-5400 for Class 2 equipment, and the maximum operating temperature for decoders shall be in accordance with MIL-E-5400 for Class 3 equipment.

3.3.8 Warm-Up Time - Parallel, frequency-coded firing systems shall be capable of operating with less than one second warm-up time.

3.3.9 Input Electrical Power -

3.3.9.1 Operating Power - Parallel, frequency-coded firing systems shall meet all applicable requirements of MIL-STD-704 without being damaged, and shall provide specified performance when supplied with power having characteristics as defined in MIL-STD-704, Category B (30 volts maximum, 21 volts minimum). Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment. The transient 28 volt d-c requirements of MIL-STD-704 shall be tested in accordance with 127AS130. The power required from the aircraft power bus shall not exceed 28 watts. Power is required intermittently by the encoder during weapons release. The decoders function as switches in the weapon-release circuits; therefore, no power is required from the aircraft power bus for decoder operation. Vendors testing decoders shall have available a regulated, d-c power supply, to provide the load currents specified on the individual decoder drawing.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements of satisfactory performance apply to performance under both standard and extreme service and input-power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

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3.4.1 Parallel Frequency-Coded Firing System Performance - Parallel, frequency-coded firing systems shall perform in accordance with the following:

(1) The encoder shall be capable of operating as many as 15 decoders simultaneously within any operating and loading condition specified on the individual decoder drawing.

(2) The peak surge voltage resulting from the decoder load interruption shall not exceed 60 volts at the decoder d-c voltage output under normal system operating conditions.

(3) The operating frequencies of the encoder and decoder shall be 7.00 ± 0.24 KHz, and should be in accordance with controlling procurement drawings.

(4) The system shall be capable of controlling the firing of weapon electro-explosive devices. Loads equivalent to aircraft weapons firing devices and wiring are specified on the individual decoder drawing.

3.4.1.1 Parallel System Decoder Performance - The output and load of decoders used in parallel, frequency-coded firing systems shall be as specified on the individual decoder drawings.

3.4.1.2 Parallel System Operating Sequence and Cycle of Operation - The operating sequence and cycle of operation for a parallel frequency-coded firing system shall be in accordance with the following:

(1) An operating sequence, for test purposes, shall be a 1 to 15 second application of input voltage to the encoder. Simultaneously with the application of input voltage to the encoder, the d-c voltage input to the decoder shall be applied for 1 to 15 seconds; this voltage shall be removed at the end of the test period.

(2) The system shall be operated six times per minute for life and reliability assurance testing.

3.5 Detail Requirements -

3.5.1 Encoders and Decoders - Encoders and decoders for parallel, frequency-coded firing systems shall be manufactured to meet the requirements of the drawings listed in Data List 154AS100.

3.5.2 Connectors - Encoder and decoder external connectors shall mate with the applicable mating connector specified on the individual encoder and decoder drawings.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests, to determine compliance with all applicable requirements.

(1) Preproduction (First Article) Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor on four equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted, so that a Government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications, and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of MIL-STD-810 and as modified in table I. The equipment shall operate satisfactorily under any of the environmental service conditions specified in table I. The equipment shall be operated at the beginning and conclusion of each service condition, to verify whether a failure occurred during the service condition. Additional operation during testing shall be made when specified by the test method of MIL-STD-810. The outputs shall be monitored during operation, to verify that output voltages are present only when the prescribed inputs are present. The operating sequence and cycle of operation shall be as specified in 3.4.1.2.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction samples shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction samples. Prefabrication of production equipment prior to the approval of the preproduction samples is at the contractor's own risk. The approved preproduction samples shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction samples shall not be deliverable under the contract.

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Table I. Environmental Tests

Environmental Service Condition	Test Method of MIL-STD-810	Notes
Temperature-altitude	Method 504 (Class 2 equipment)	For testing encoders.
Temperature-altitude	Method 504 (Class 3 equipment)	For testing decoders. MAX altitude shall be 70,000 ft.
Temperature-shock	Method 503	Equipment shall be operating
Humidity	Method 507 Procedure I	
Vibration	Method 514.1 Curve AT	From 5 to 15 Hz. The double amplitude shall be ± 0.2 inch instead of + 0.1 inch.
Shock	Method 516 Procedure I	Half-sine-wave shock pulse, 11 milliseconds duration. Basic Design - 15g. Crash Safety - 30g. Apply d-c power during test and monitor for inadvertent firing.
Salt fog	Method 509	
Fungus	Method 508	Fungus testing is not required on units with all components except for the connector mounted in a sealed metal enclosure.
Sand and Dust	Method 510	Sand and Dust testing is not required on units with all components except for the connector mounted in a sealed metal enclosure.
Explosive atmosphere	Method 511 Procedure I	Explosive atmosphere testing is not required on units with all components except for the connector mounted in a sealed metal container.

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4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection, and for equipment already accepted by the Government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - Four of the first 25 production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on the equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to ensure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality, and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests, the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted, when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - Upon completion of the initial production test if internal electrical parts are not worn or damaged, the equipment shall be reworked by the contractor by replacing all wear or damaged items located external to the equipment housing. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be subject to the approval of the Government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the furnished product. Acceptance tests shall consist of the following:

(1) Individual Tests

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(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests.

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run-In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully, to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to ensure satisfactory equipment operation.

4.4.1.3 Manufacturing Run-In Test - Each equipment shall be operated in accordance with the operating sequence and cycle of operation of 3.4.1.2 under the conditions specified herein for a period of 10 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. No adjustments will be permitted during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least 10 hours of operation.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of ±3g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 10-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the 10-hour period of operation following the 10 minute vibration period, the equipment shall be electrically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test restarted, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective

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action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 10 hour period specified above may be composed of two 5-hour periods, to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following:

Number of Equipments Offered for Acceptance	Number to be Selected for Sampling Test
First 25	4#
Next 50 or fraction thereof	1
Next 75 or fraction thereof	1
Next 100 or fraction thereof	1
	1 for each additional 200 or fraction thereof

Sampling Tests are not required when Reliability Assurance Tests are conducted.

#NOTE: If by contract action the Initial Production Test is deleted, a Sample Test shall also be conducted on four equipments from the first 25 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to ensure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested, and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run-in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling)Phase shall be conducted. Simulated firings shall be made at the rate of 6 per minute for each channel. Encoders shall be loaded in a manner to simulate loads of 125, 500, 1000 and 9.5K ohms. At least 25,000 firings shall be made with each of the 4 simulated loads. Decoders shall be operated with loads

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at the duty cycles specified on the equipment drawing. At least 10,000 firings shall be made with each of the load conditions at the duty cycle specified. The input signal to the decoder shall be within the limits of the drawing requirements for the outputs of the encoders. Decoder failures shall be defined as the occurrence of an unwanted firing voltage or lack of a wanted firing voltage at the decoder load. Encoder failures shall be defined as the presence of an out of tolerance output voltage or frequency at the simulated load.

4.4.3.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three equipments shall be tested as outlined in MIL-STD-781, under the Section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in table 5 of MIL-STD-781. For the Qualification Phase, Test Level F shall be used, except equipment shall be operated continuously and a simulated altitude of 70,000 feet shall be applied for 1 hour minimum during each low-temperature period. The Accept-Reject Criteria of table II shall be used.

Table II. Accept-Reject Criteria for Qualification Phase

No. of Failures	Number of Operations to	
	Reject	Accept
0	N/A	238,000
1	N/A	312,000
2	N/A	388,000
3	38,000	462,000
4	112,000	537,000
5	188,000	612,000
6	262,000	687,000
7	337,000	761,000
8	413,000	837,000
9	487,000	912,000
10	562,000	986,000
11	637,000	1,062,000
12	712,000	1,112,000
13	786,000	1,112,000

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14	870,000	1,112,000
15	936,000	1,112,000
16	1,112,000	N/A

4.4.3.2 Reliability Production Acceptance (Sampling) - Samples of the production equipment shall be tested as outlined in MIL-STD-781 (as modified herein), under the Section entitled "Sampling Phase of Production Reliability Tests." For the Sampling Phase (or All-Equipment Test), Test Level F shall be used except equipment shall be operated continuously and a simulated altitude of 70,000 feet shall be applied for one hour minimum during each low-temperature period. The accept-reject criteria of table III shall be used.

Table III. Accept-Reject Criteria for Production Phase

No. of Failures	Number of Operations To	
	Reject	Accept
0	N/A	151,000
1	N/A	226,000
2	38,000	301,000
3	112,000	376,000
4	187,000	450,000
5	262,000	526,000
6	337,000	526,000
7	411,000	526,000
8	526,000	N/A

4.4.3.2.1 Lot Size for Sampling Phase - The equipments constructed during one month shall be one lot. The Accept-Reject criteria in table 3 shall be used to determine the number of firings required for each lot. Unless otherwise specified in the contract or order, no equipments shall be shipped until an Accept decision is reached under each lot.

4.4.3.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

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4.4.4 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment, and to ensure adequate quality control. The equipments selected for special tests may be equipments previously subjected to the sampling tests or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity).

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practicable, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

(5) When a reject decision is reached on a lot as a result of isolated or radom type failures, the entire lot shall be subjected to the Manufacturing Run-In-Test of 4.4.1.3, following which new samples shall be tested in accordance with 4.4.3.2.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall continue until 129,000 operations have been accomplished. Of the 129,000 operations, 43,000 may have been made at standard conditions (production and acceptance tests apply) and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test shall be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 100,000 operations.

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Number of Equipments Offered for Acceptance	Number to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room - for first 40,000 operations. In accordance with table 1 for the balance of the life test.
Altitude	Test Facility ground (0-5000 ft.)
Humidity	Room or chamber ambient
D. C. Voltage	27.5 ± 0.5 volts
Operating Sequence and Cycle of Operation	As specified in 3.4.1.2

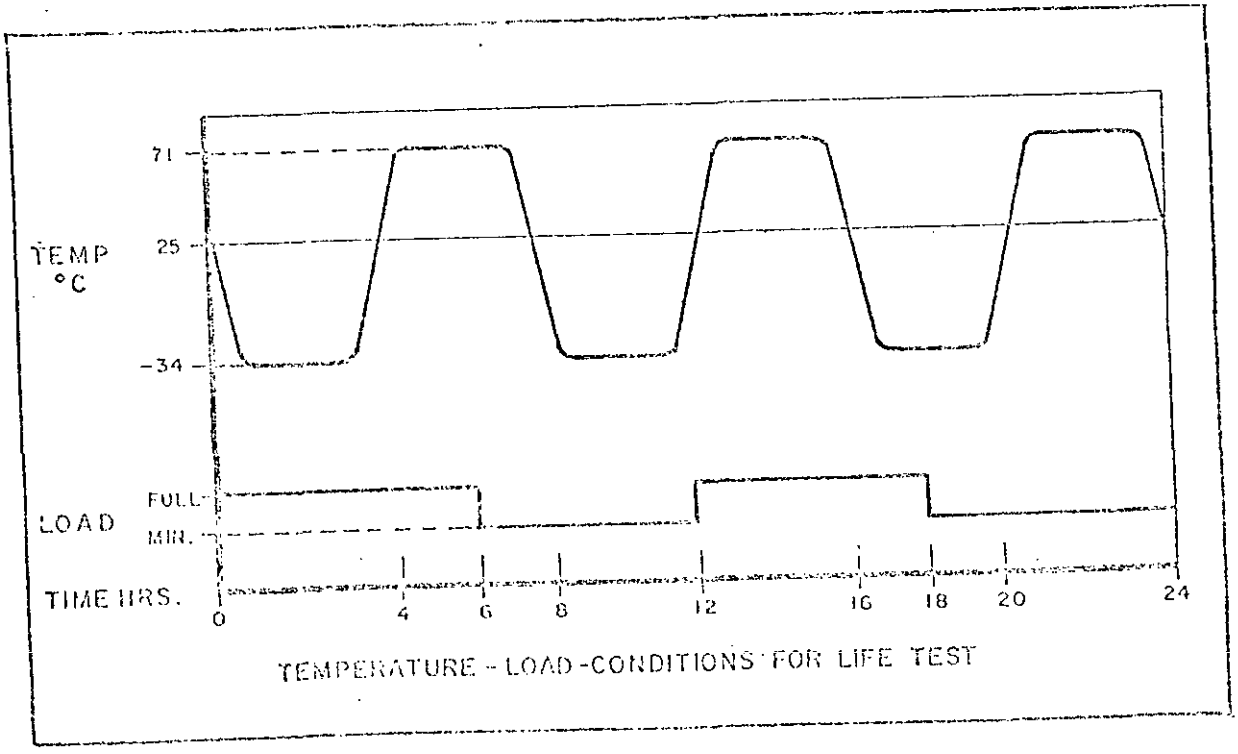


Figure 1. Temperature-Load-Conditions for Life Test

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4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximate 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance. (see 4.3.4)

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

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4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - A frequency-coded firing system is installed within Navy aircraft weapons release systems to preclude an accidental weapons release caused by the accidental application of voltage onto a firing line or the electro-magnetic coupling of r-f energy onto a firing line.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification

(2) Responsibility for Inspection (See 4.1)

(3) Selection of applicable levels of packaging and packing (See 5.1)

(4) The activity that is to perform the preproduction and reliability assurance testing. (Government approved laboratories having special test equipment and approved test procedure shall be used when practicable).

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears that a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

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